

AIML SESSION 04 ASSIGNMENT SUBMISSION

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Domain/subject name: AIML (Artificial Intelligence & Machine Learning)

College and Branch name: Government Polytechnic College Washim (IF)

Task 1. Lambda Function

Write a lambda function named `square` that takes a number `x` and returns its square. Call the lambda function with input from the user and print the result. Also write the same logic using a normal `def` function and compare both in comments.

Program Code:

```
# Lambda Function

square = lambda x: x * x
number = float(input("Enter a any number: "))

print("Square using lambda:", square(number))

# Normal Function

def square_def(x):
    return x * x

print("Square using normal function:", square_def(number))
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q1.py"
Enter a any number: 2
Square using lambda: 4.0
Square using normal function: 4.0
○ > █
```

Task 2: Lambda with Multiple Arguments

Create a lambda function `add_three(a, b, c)` that returns the sum of three numbers. Take three numbers as input from the user and print their sum using this lambda function.

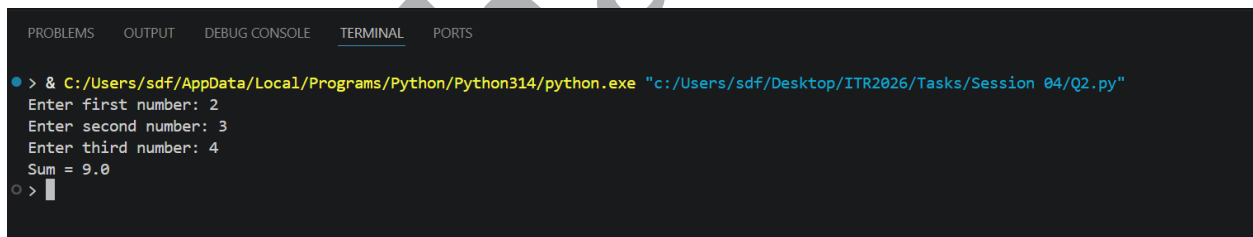
Program Code:

```
add_three = lambda a, b, c: a + b + c

a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

print("Sum =", add_three(a, b, c))
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q2.py"
Enter first number: 2
Enter second number: 3
Enter third number: 4
Sum = 9.0
○ > |
```

Task 3: . For Loop Revision

Write a program using a for loop and range() to: a) Print all numbers from 1 to 25. b) Print the sum of all numbers from 1 to 20. c) Print the table of 5 1 to 5 10. Add comments explaining when to use for loop.

Program Code:

For loop is used for repeating a tasks a specific number of times. It is used when the number of iterations is known beforehand. The syntax of a for loop in Python is as follows:

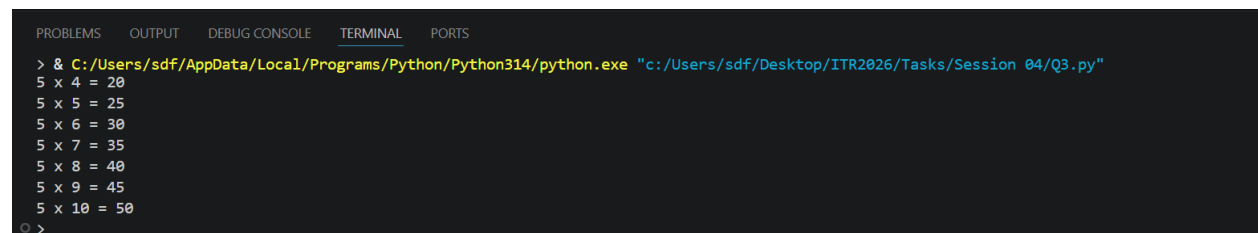
```
print("Numbers from 1 to 25:")
for i in range(1, 26):
    print(i)
tot = 0

for i in range(1, 21):
    tot += i

print("Sum of numbers from 1 to 20 =", tot)
print("Table of 5:")

for i in range(1, 11):
    print(f"5 x {i} = {5 * i}")
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q3.py"
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
○ >
```

Task 4: . While Loop Revision

Write a program using a while loop that: Repeatedly asks the user to enter a positive number. Keeps adding the numbers to a running total. Stops when the user enters 0 or a negative number. Finally prints the total sum.

Program Code:

```
# Initialize sum variable
total = 0

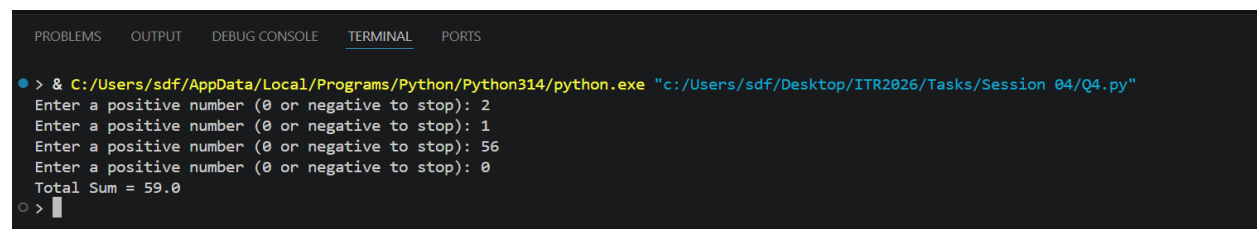
# Loop until user enters non-positive number
while True:
    # Get number from user
    value = float(input("Enter a positive number (0 or negative to stop): "))

    # Exit loop if number is not positive
    if value <= 0:
        break

    # Add to running total
    total += value

# Display final sum
print("Total Sum =", total)
```

OUTPUT Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
• > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q4.py"
Enter a positive number (0 or negative to stop): 2
Enter a positive number (0 or negative to stop): 1
Enter a positive number (0 or negative to stop): 56
Enter a positive number (0 or negative to stop): 0
Total Sum = 59.0
○ > █
```

Task 5: Math Module

Write a program that imports the math module and does the following:
Takes a number as input. Prints its square root using `math.sqrt()`. Prints its factorial using `math.factorial()` (for positive integers only). Prints the ceiling and floor value using `math.ceil()` and `math.floor()`. Add comments about the usefulness of the math module.

Program Code:

```
import math

# Get numeric input from user
value = float(input("Enter a number: "))

# Calculate square root (only for non-negative values)
if value < 0:
    print("Square Root = Not defined for negative numbers")
else:
    print("Square Root =", math.sqrt(value))

# Calculate and display ceiling and floor values
print("Ceiling Value =", math.ceil(value))
print("Floor Value =", math.floor(value))

# Calculate factorial (only for non-negative integers)
if value >= 0 and value.is_integer():
    print("Factorial =", math.factorial(int(value)))
else:
    print("Factorial can only be calculated for non-negative integers.")
```

OUTPUT Screenshot:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q5.py"
Enter a number: 12
Square Root = 3.4641016151377544
Ceiling Value = 12
Floor Value = 12
Factorial = 479001600
> 
```

Vaibhav Kate

Task 6: Random Module

Write a program using the random module that: Generates and prints 5 random integers between 1 and 100. Generates and prints one random number between 50 and 150. Prints a random floating point number between 0 and 1. Use different functions from the random module.

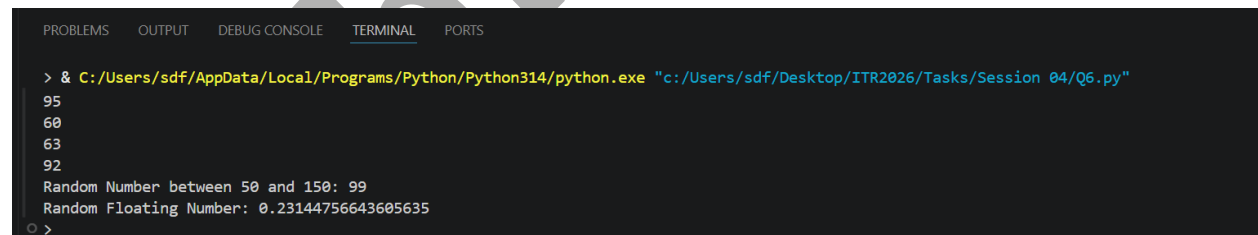
Program Code:

```
import random

print("five random integers:")

for i in range(5):
    print(random.randint(1, 100))
print("Random Number between 50 and 150:", random.randint(50, 150))
print("Random Floating Number:", random.random())
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q6.py"
95
60
63
92
Random Number between 50 and 150: 99
Random Floating Number: 0.23144756643605635
○ >
```


Task 7: Import Methods

Demonstrate different ways of importing modules: Import the entire math module using `import math` and use `math.pow(2, 4)`. Import only `sqrt` from math using `from math import sqrt`. Import math with an alias as `m` and use `m.factorial(5)`. Print the results with clear messages.

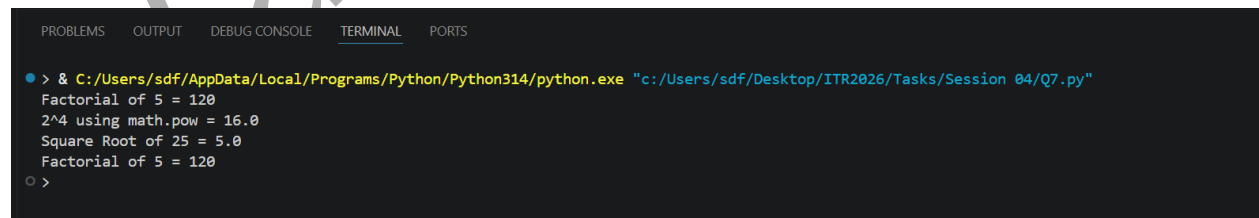
Program Code:

```
# Method 1
import math
print("Factorial of 5 =", math.factorial(5))
print("2^4 using math.pow =", math.pow(2, 4))

# Method 2
from math import sqrt
print("Square Root of 25 =", sqrt(25))

# Method 3
import math as m
print("Factorial of 5 =", m.factorial(5))
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q7.py"
Factorial of 5 = 120
2^4 using math.pow = 16.0
Square Root of 25 = 5.0
Factorial of 5 = 120
>
```

Task 8: Custom Module

Create a custom module file named `mymodule.py` that contains: A function `greet(name)` that prints "Hello [name], Welcome to Python Class!" A function `calculate_power(base, exp)` that returns base raised to exponent. Then write a main program that imports this module and uses both functions.

Program Code:

```
# Main program that imports and uses the custom module
```

```
import mymodule
```

```
# Using greet() function
```

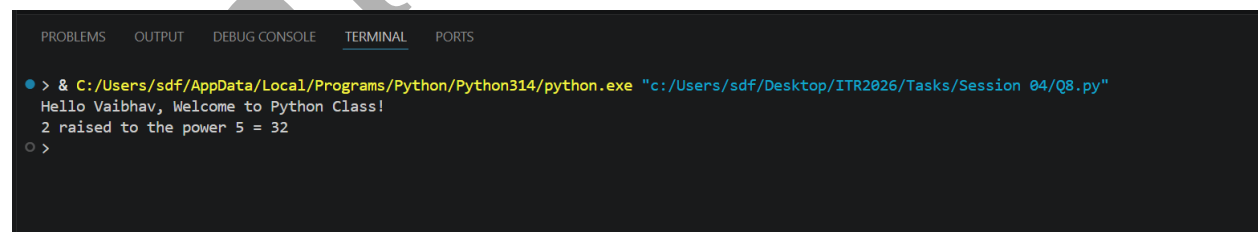
```
mymodule.greet("Vaibhav")
```

```
# Using calculate_power() function
```

```
result = mymodule.calculate_power(2, 5)
```

```
print("2 raised to the power 5 =", result)
```

OUTPUT Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q8.py"
Hello Vaibhav, Welcome to Python Class!
2 raised to the power 5 = 32
>
```

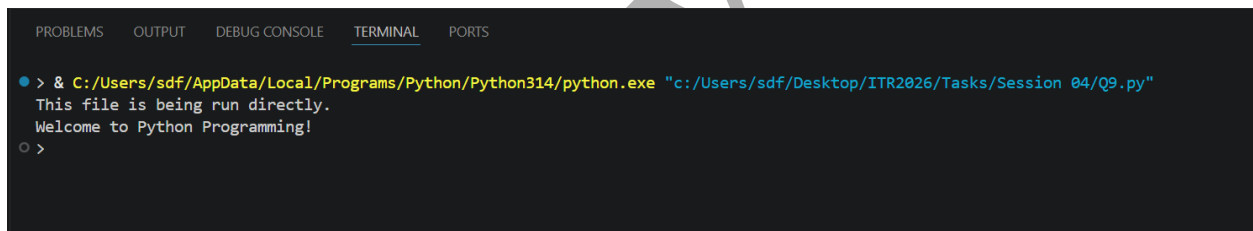
Task 9: (name == "main")

Write a program that: a) Creates a small function and demonstrates the use of `if __name__ == "__main__":` so that some code runs only when the file is executed directly.

Program Code:

```
def greet():  
    print("Welcome to Python Programming!")  
  
if __name__ == "__main__":  
    print("This file is being run directly.")  
    greet()
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q9.py"  
This file is being run directly.  
Welcome to Python Programming!  
○ >
```

Task 10: Mini Project –

Combined Concepts) Create a Simple Math Utility Program using functions, lambda, loops and modules: Write the following: A lambda function to calculate square of a number. A normal function using math module to calculate power. Use a while loop to repeatedly ask the user for a number and choice Square / Power / Random Number). Use random module to generate a random number when chosen. Continue until the user chooses to exit. Use proper functions and comments. 3 Session 04 (AIML) - Assignment Questions

Program Code:

```
# Math Utility Program - Combined Concepts
import math
import random

# Lambda function for square
square = lambda x: x ** 2

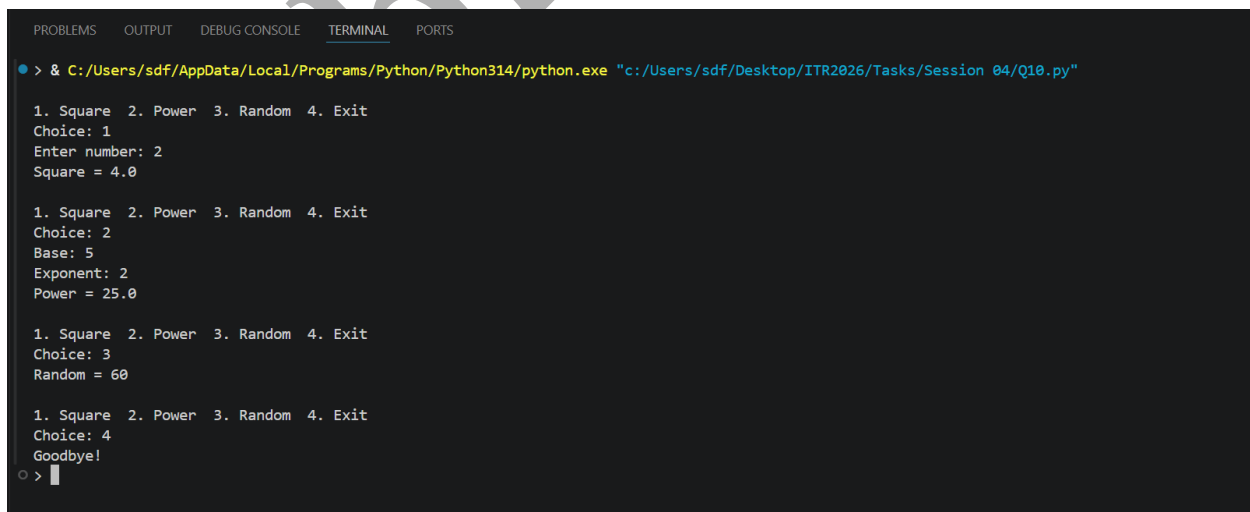
# Function for power calculation
def calculate_power(base, exp):
    return math.pow(base, exp)

# Function for random number
def generate_random():
    return random.randint(1, 100)

# Main program
while True:
    print("\n1. Square 2. Power 3. Random 4. Exit")
    choice = input("Choice: ")
```

```
if choice == '1':
    num = float(input("Enter number: "))
    print(f"Square = {square(num)}")
elif choice == '2':
    base = float(input("Base: "))
    exp = float(input("Exponent: "))
    print(f"Power = {calculate_power(base, exp)}")
elif choice == '3':
    print(f"Random = {generate_random()}")
elif choice == '4':
    print("Goodbye!")
    break
else:
    print("Invalid choice!")
```

OUTPUT Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 04/Q10.py"

1. Square 2. Power 3. Random 4. Exit
Choice: 1
Enter number: 2
Square = 4.0

1. Square 2. Power 3. Random 4. Exit
Choice: 2
Base: 5
Exponent: 2
Power = 25.0

1. Square 2. Power 3. Random 4. Exit
Choice: 3
Random = 60

1. Square 2. Power 3. Random 4. Exit
Choice: 4
Goodbye!
>
```